TASK 4:

```
#include <iostream>
using namespace std;

int main()
{
   int n;
   int *arr;
   int sum = 0;
   float avg = 0.0;

   cout << "Enter size of an array: ";
   cin >> n;

   arr = new int[n];

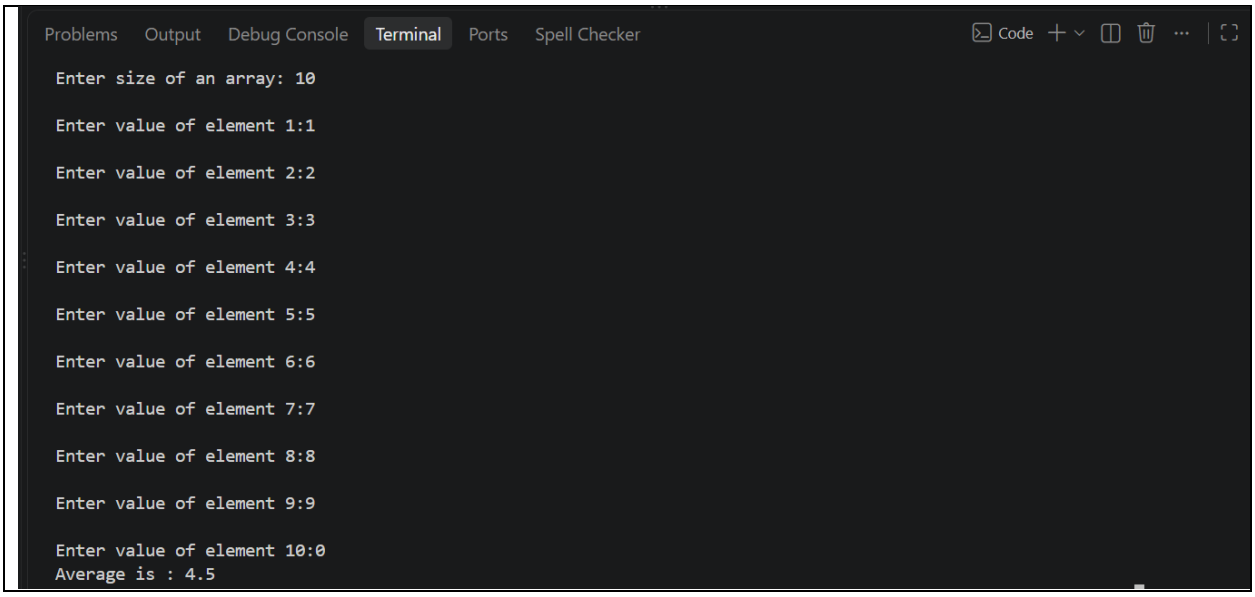
   for (int i = 0; i < n; i++)
   {
      cout << "\nEnter value of element " << i + 1 << ":";
      cin >> *(arr + i);

      sum = sum + *(arr + i);
   }

   avg = (float)sum / n;

   cout << "Average is : " << avg;

   delete[] arr;
}
```



TASK 5:

```
#include<iostream>
using namespace std;

class Product
{
  string product_name;
  int *quantity;

  public:
  Product(string name, int qty)
  {
    product_name=name;
    quantity=new int;
    *quantity=qty;
  }

  Product(const Product& orig)
  {
    product_name=orig.product_name;
    quantity=new int;
    *quantity=*(orig.quantity);
  }

  Product& operator=(const Product& orig)
  {
```

```
    if (this != &orig)
    {
      delete quantity;

      product_name=orig.product_name;

      quantity = new int;

      *quantity=*(orig.quantity);
    }

    return *this;
  }

  void display()
  {
    cout<<"Product name : "<<product_name<<endl;
    cout<<"Quantity : "<<*quantity<<endl;
  }

  void setQuantity(int qty)
  {
    *quantity=qty;
  }

  ~Product()
  {
    delete quantity;
  }
};

int main()
{
  Product obj1("Laptop", 5);

  cout << "Original Object:\n";
  obj1.display();

  // Copy constructor
  Product obj2(obj1);

  cout << "\nObject 2 after copying Object 1:\n";
  obj2.display();
```

```
    // Modify copied object
    obj2.setQuantity(10);

    cout << "\nAfter modifying Object 2:\n";

    cout << "Object 1:\n";
    obj1.display();

    cout << "\nObject 2:\n";
    obj2.display();

    // Assignment operator
    Product obj3("Mobile", 2);

    cout << "\nObject 3 before assignment:\n";
    obj3.display();

    obj3 = obj1;

    cout << "\nObject 3 after assigning Object 1 to Object 3:\n";
    obj3.display();

    return 0;
}
```

